CHAPTER I

THE PROBLEM AND ITS BACKGROUND

INTRODUCTION

Technology has continuously evolved and has significantly influenced how individuals communicate, work, and learn. Innovations in digital technologies have enabled organizations and institutions to improve efficiency, accessibility, and the quality of services provided to users. In the education sector, the integration of technological solutions has transformed traditional learning environments into more flexible and data-driven systems that support both students and educators. These technological advancements have led to the development of digital learning platforms, intelligent educational systems, and online tutoring environments that allow learners to access academic support beyond the limitations of the traditional classroom.

Artificial intelligence has played an important role in enhancing modern educational technologies. One of the significant developments in this field is the emergence of Intelligent Tutoring Systems (ITS), which aim to simulate the guidance provided by human tutors using computational models and artificial intelligence techniques. According to Rodrigues, Pinto, and Gonçalves, artificial intelligence technologies are reshaping educational environments through intelligent tutoring systems that enable personalized instruction and learning support for students [1]. These systems analyze learner behavior and performance data in order to provide adaptive feedback and instructional assistance that improves the learning experience.

Another important innovation in educational technology is the use of adaptive learning systems. Adaptive learning technologies are designed to adjust learning content and instructional strategies according to the performance and behavior of learners. Research conducted by Song indicates that adaptive learning systems have been widely studied for their potential to improve students' academic achievement, engagement, and skills development [2]. Through the analysis of learner performance, adaptive learning platforms are capable of delivering personalized learning paths that help students better understand academic concepts.

In addition to adaptive learning technologies, modern educational systems also utilize learning analytics to support academic monitoring and decision-making. Learning analytics refers to the collection and analysis of educational data in order to understand and improve the learning process. According to Wang, Lai, and Huang, learning analytics has emerged as an important tool for deriving insights from student learning behaviors and identifying patterns that can improve learning outcomes [3]. By transforming learning data into meaningful information, learning analytics allows educators to monitor student progress and identify learners who may require additional academic support.

Another technology that supports personalized learning environments is the use of educational recommender systems. Recommender systems analyze user preferences, learning styles, and behavioral data in order to suggest relevant learning resources or instructional support. Educational recommendation systems have been widely applied in online learning platforms to provide personalized recommendations that align with learners' academic needs and preferences [4]. These systems improve learning efficiency by connecting students with appropriate educational resources and instructional assistance.

Despite the availability of these technologies, many students still experience difficulties in accessing tutoring services that match their academic needs. Traditional tutoring arrangements often rely on manual coordination, personal referrals, or limited tutor availability. As a result, students may struggle to find tutors who specialize in particular subjects, while tutors may encounter difficulties reaching learners who require their expertise. In many cases, tutoring sessions are conducted without proper monitoring systems that track student learning progress, making it difficult for parents and educators to evaluate whether tutoring sessions effectively improve student performance.

Because of these limitations, there is a growing need for digital platforms that can efficiently manage tutoring services while providing tools for monitoring student learning progress. Integrating intelligent tutoring technologies, recommendation systems, and learning analytics into a single platform can significantly improve the accessibility and effectiveness of tutoring services.

In response to these challenges, the researchers propose the development of Tutorly: An Intelligent Tutor Matching and Academic Progress Monitoring System. The proposed system aims to provide a centralized web-based platform that connects students, tutors, and parents while facilitating intelligent tutor–student matching based on compatibility factors such as subject expertise, availability, learning difficulty, tutor ratings, and budget compatibility. In addition, the system will include features that allow tutors to record tutoring sessions, administer quizzes and assessments, and generate progress reports that monitor the academic development of students.

Through the integration of intelligent tutoring systems, recommendation algorithms, and learning analytics, the proposed system aims to improve the efficiency of tutoring services while providing a structured approach to monitoring academic progress. By leveraging modern educational technologies, the system is expected to enhance communication among stakeholders and contribute to improving the overall tutoring experience for students.

Background of the Study

The use of digital technologies in education has changed the way students learn in traditional settings, allowing them to get help with their studies outside of the classroom. In the last ten years, online tutoring platforms have become popular ways for students to learn in a way that works for them. These platforms let students connect with qualified tutors no matter where they are. These platforms have gotten even better thanks to improvements in artificial intelligence (AI), adaptive learning systems, and learning analytics. They now offer personalized teaching methods, the ability to track progress, and educational insights based on data. Rodrigues et al. stress that intelligent tutoring systems powered by AI can make learning more effective by tailoring content and feedback to each student's needs [1]. Singh et al. also say that intelligent tutoring systems are using machine learning more and more to improve recommendations and teaching choices [2].

Adaptive tutoring systems can make learning more personal by figuring out what each student is good at, what they need to work on, and how they learn best. Patil et al. show that these systems can change how they teach to make personalized learning paths, which is especially helpful for students who are having trouble in school [3]. Furthermore, the utilization of learning analytics furnishes educators and institutions with actionable insights regarding student performance, engagement, and progress, facilitating data-driven educational interventions [4]. Recommendation systems make personalized learning even better by matching the learner's needs and preferences with the right educational content and tutor expertise [5].

Even though technology has come a long way, many communities, including Nasugbu, Batangas, still have trouble finding organized and trustworthy tutoring services. Students usually use informal ways to find tutors, like talking to friends, getting referrals from school, going to a tutorial center, or joining a social media group. These methods don't always work, don't have a clear plan, and don't guarantee that the tutor will be good, that the schedule will work, or that progress will be tracked. Some common problems are having tutors who don't match, not being able to verify credentials, poor communication, not having a central platform, and using third-party tools for quizzes and tests that don't work with a single system. These gaps show that there is a need for a smart, centralized, and localized platform that meets the educational needs of each community.

Recent literature bolsters the adoption of AI-driven, adaptive, and data-informed tutoring systems in K-12 education. Létourneau et al. assert that AI-driven systems can identify learning challenges, modify instructional approaches, and track student advancement, thereby improving individualized support in foundational education [6]. Latif et al. point out that intelligent tutoring systems have changed to combine AI-based classification with pedagogical design. This makes it possible to create more accurate student profiles and adaptive learning interventions [7]. Mustafa et al. stress how important AI is to modern schools, pointing out how it can help with personalized instruction, tracking progress, and learning analytics [8]. Li et al. and Fernández-Herrero also show that advanced AI algorithms, like graph-based recommendation models and effective tutoring systems, can improve the matching of tutors and students, keep track of learning outcomes, and make students more interested in learning [9][10].

These results support the creation of Tutorly: An Intelligent Tutor Matching and Academic Progress Monitoring System for Learners in Nasugbu, Batangas. This is a web-based, mobile-responsive platform that aims to fill the gaps in current tutoring services. The system has AI-based tutor recommendations, profiles of learning difficulties, progress tracking, structured communication, tutor verification and rating, and data analytics. Tutorly wants to make tutoring services easier to get to, more reliable, and more effective by using smart technologies. It also wants to give local education offices like the Youth Development Office useful information. The platform helps students by finding tutors who are a good fit for them and helping them with their learning problems. It also makes tutors more accountable and helps schools make decisions about how to help students based on data.

In conclusion, the proposed system fixes problems with tutoring services in Nasugbu, fits in with current trends in AI-based intelligent tutoring, and is community-focused, encouraging personalized learning, keeping track of academic progress, and providing good educational support.

OBJECTIVES OF THE STUDY

The primary objective of this study is to design and develop Tutorly: An Intelligent Tutor Matching and Academic Progress Monitoring System for learners in Nasugbu, Batangas that utilizes learning difficulty profiling and a rule based compatibility algorithm to improve tutor–student matching, support academic progress monitoring, and provide an organized platform for communication between students, tutors, and parents.

To achieve this general goal, the study sets the following specific objectives:

1. To develop a centralized platform that connects students, parents, and tutors in Nasugbu, Batangas to improve access to tutoring services.

2. To design a learning difficulty profiling module that identifies students' academic challenges using diagnostic questions, quizzes, tutor assessments, and academic records.
3. To implement an intelligent tutor recommendation system that suggests compatible tutors based on subject expertise, availability, location, learning difficulty, ratings, and budget compatibility.
4. To develop a student progress monitoring system that records tutoring sessions, quiz and examination scores, tutor evaluations, attendance, and visual progress reports.
5. To provide structured communication and assessment tools, including messaging features and an online quiz and examination module to support tutoring sessions.
6. To implement a tutor verification, rating, and secure payment system to ensure platform credibility and reliable transaction management.
7. To design an administrative dashboard that allows administrators to manage accounts, monitor system activities, and generate analytical reports.

SIGNIFICANCE OF THE STUDY

The implementation of "Tutorly: An Intelligent Tutor Matching and Academic Progress Monitoring System" is designed to improve the accessibility and organization of tutoring services through a centralized digital platform. The system aims to facilitate efficient tutor-student matching, monitor academic progress, and provide structured communication among students, tutors, and parents. The proposed system is expected to benefit several stakeholders, including students, tutors, parents, educational institutions, and future researchers.

Students Tutorly provides students with easier access to tutoring services by allowing them to search and connect with tutors who specialize in the subjects they need help with. Through the intelligent tutor recommendation feature, students can find tutors based on compatibility factors such as subject expertise, availability, and learning difficulty. The system also allows students to monitor their academic progress through recorded tutoring sessions, quizzes, and performance reports, helping them better understand their strengths and areas that need improvement.

Tutors The system provides tutors with a structured platform where they can manage tutoring sessions, monitor student progress, and provide feedback effectively. Tutorly allows tutors to track student performance through assessment results and progress reports, enabling them to adjust their teaching strategies according to the needs of the learners. In addition, the system helps tutors expand their reach by connecting them with students who require their expertise.

Parents or Guardians Tutorly allows parents to stay informed about their child's academic progress by providing access to performance reports, tutoring session records, and evaluation results. This enables parents to monitor the effectiveness of tutoring sessions and support their child's academic development. The system also promotes transparency by allowing parents to observe the progress and improvements of their children over time.

Educational Institutions and Tutoring Organizations Educational institutions and tutoring service providers can benefit from Tutorly by having a centralized platform that organizes tutor profiles, student records, and tutoring session data. The system allows administrators to monitor tutoring activities, generate reports, and analyze learning performance trends. This helps institutions improve the management of tutoring services while ensuring that students receive appropriate academic support.

Future Researchers. The study may serve as a reference for future researchers who are interested in developing intelligent tutoring platforms, recommendation systems, or learning analytics technologies in the field of education. It may also contribute to further studies related to personalized learning systems and academic progress monitoring platforms

SCOPE AND LIMITATION OF THE STUDY

The study concentrates on the creation of Tutorly: An Intelligent Tutor Matching and Academic Progress Monitoring System for Learners in Nasugbu, Batangas, a web-based platform intended for private tutors, students, and parents to obtain structured tutoring services within the municipality. The system's goal is to make it easier for students to find the right tutors by giving them a central place to find tutors based on their academic needs, subject requirements, availability, and other factors that make them a good match. The study aims to facilitate more effective tutor-student matching and offer tools for tracking academic progress.

The proposed system will have a number of modules that help with the tutoring process. The Learning Difficulty Profiling module will let students answer structured questions that help find common learning problems like reading comprehension, solving math problems, studying, and staying focused. The system's Intelligent Tutor Recommendation Engine will use the information it collects. This engine uses a rule-based compatibility algorithm that looks at the tutor's expertise, subject specialization, schedule availability, location, and budget. The system will use these criteria to make suggestions with compatibility scores to help parents and students pick the right tutor.

The platform will also have a student progress tracking system that lets tutors write down what they talked about, what they worked on, and how the student did after each session. These records will be organized and shown in visual summaries to help parents and students keep track of how their grades are getting better over time. The system also lets users register and verify their accounts, has a calendar-based scheduling system for booking tutoring sessions and showing session schedules, and has a structured messaging module that lets tutors and students talk to each other while keeping a record of their conversations on the platform. The system will also have a tutor verification and rating feature. Tutors will have to upload proof of their qualifications for administrative approval, and students and parents will be able to leave feedback after each session. Administrators will be able to manage user accounts, check tutor credentials, keep an eye on system activities, and make reports about system use and tutoring demand with an Administrative Dashboard. The platform will also have an online quiz and examination feature that lets tutors make tests and a payment management module that shows the cost of tutoring and keeps track of payments.

The system will be built as a web-based app that is mobile-device responsive, so people can use it on their desktop, laptop, or mobile browser as long as they have an internet connection. A dual-core processor, at least 4GB of RAM, enough space for database records, and a stable internet connection are the bare minimum hardware requirements. For server-side scripting, the development environment will use PHP; for database management, it will use MySQL; for interface design, it will use HTML and CSS; for interactive features, it will use JavaScript; and for some processing tasks, it will use Python. XAMPP will be used as the development server for the system, and it will be tested and developed locally. People will be able to access it through modern web browsers like Google Chrome or Microsoft Edge.

The Agile Systems Development Life Cycle (SDLC) method will be used to make the system. We chose Agile because it lets us develop in stages and change system modules as needed. This method also makes it possible to get feedback from potential users during testing, which helps make sure that the system features meet the needs of the students, parents, and tutors.

This study will employ a qualitative research design to comprehend current tutoring practices, the challenges faced by tutors and students, and the necessary requirements for creating an effective tutor matching platform. Interviews, observations, and document reviews will be used to collect information that is useful for system design. The main people who will take part in the study are private tutors in Nasugbu, Batangas, who will share their thoughts on tutoring methods and system needs. For purposive sampling, people are chosen based on their experience and involvement in tutoring activities. There should be about five to ten tutors who can give useful information for the system's development.

Despite its intended contributions, the study has several limitations. The system was made just for basic education students in the Municipality of Nasugbu, Batangas. It may need to be changed before it can be used in other places or for higher levels of education. The tutor recommendation feature uses a rule-based compatibility algorithm, which means that the suggestions are based on rules that have already been set up, not on advanced machine learning models. The learning difficulty profiling module uses structured questionnaires to help match tutors, but it is not a professional educational or psychological diagnostic test. Finally, the study mainly looks at the system's design, development, and prototype implementation. The long-term evaluation of its impact on student academic performance is beyond the scope of this research.

DEFINITION OF TERMS

To avoid ambiguity, here are the following terms used in this paper:

Academic Progress Monitoring The process of tracking and evaluating a student’s academic performance over time using assessments, tutoring records, quizzes, and performance reports to determine learning improvement.

<u>Adaptive Learning System</u> A technology-based learning system that adjusts instructional content, difficulty level, and learning pathways based on the learner's performance, behavior, and academic needs.

<u>Administrator Dashboard</u> A system interface used by administrators to manage user accounts, verify tutor credentials, monitor system activities, and generate analytical reports related to tutoring sessions and platform usage.

<u>Artificial Intelligence (AI)</u> A branch of computer science that enables computer systems to perform tasks that normally require human intelligence, such as decision-making, pattern recognition, learning from data, and problem-solving.

<u>Compatibility Algorithm</u> A rule-based computational method used to evaluate and calculate the suitability between students and tutors based on factors such as subject expertise, availability, location, ratings, and learning difficulty.

<u>Intelligent Tutor Matching</u> A system feature that automatically recommends suitable tutors to students by analyzing compatibility factors such as academic needs, tutor specialization, availability, and performance ratings.

<u>Intelligent Tutoring System (ITS)</u> A computer-based educational system that uses artificial intelligence techniques to simulate the guidance of a human tutor by

providing personalized feedback, instructional support, and learning recommendations.

Learning Analytics The process of collecting, measuring, analyzing, and interpreting educational data to understand student learning behavior and improve academic outcomes.

Learning Difficulty Profiling A system module that identifies students' academic challenges by analyzing responses from diagnostic questions, tutor evaluations, and performance data to determine the type of academic assistance required.

Online Tutoring Platform A digital environment that allows students and tutors to connect through the internet for academic assistance, tutoring sessions, communication, and learning support.

Progress Report A structured summary generated by the system that shows a student's academic performance, tutoring session results, quiz scores, and learning improvement over time.

Rule-Based System A decision-making system that operates using predefined rules or logical conditions to generate outcomes, recommendations, or system responses.

Tutor Verification System A feature that validates tutor credentials and qualifications before allowing them to provide tutoring services on the platform to ensure reliability and credibility.

Tutor Rating System A feedback mechanism that allows students and parents to evaluate tutors based on their performance, teaching effectiveness, and reliability after tutoring sessions.

Tutorly System A web-based intelligent tutor matching and academic progress monitoring platform designed to connect students, tutors, and parents while facilitating personalized tutoring services and learning analytics.

Web-Based Application A software application that runs on a web server and can be accessed through internet browsers without requiring installation on a local device.